

Interview Questions — Doctor and Patient

Dana Al Mounayer & Jwdee Alimam · Supervisor: Dr. Naila Marir

THE RULE FOR EVERY QUESTION

Every question asks about a fact — something that was done, said, measured or recorded.

No question asks for an opinion. No question asks what someone would do in future. No question asks about an app, a design or BADIRA. The purpose is to test whether the problem described in Report 1 is real in Saudi practice, and in one real pregnancy.

The seven literature themes guiding the interviews

These seven questions were run as a literature search. Each returned at least one paper, listed below with its link and what it reports. Each interview question is mapped to one of these literature themes or to a clinical guideline, so every question has a clear evidence-based reason for being included.

These literature themes are not questions that will be asked directly to participants; they are the evidence base used to design and justify the interview questions.

Literature finding → Interview question → Real doctor/patient experience → Comparison → Report 1 update

P1 · Do women who should receive aspirin actually get it?

What proportion of pregnant women identified as at high risk of preeclampsia actually receive and start low-dose aspirin before 16 weeks, and what are the barriers to uptake?

Design: 2,567 pregnancies among 2,563 women in one health system. Survey answers from the MyHealthyPregnancy app were compared against the women's own medical records. Findings: of 411 women with multiple moderate risk factors, only 46% received an aspirin recommendation from their clinician. Of 124 women with at least one high-risk factor, 27% did not recall being advised to take it. And of 90 women whose records contained documented evidence that a clinician had advised aspirin, only 57 — 63% — were aware of it. Conclusion: frequent miscommunication between women and clinicians about low-dose aspirin. The third figure is the one to quote: the recommendation was in the chart, and more than a third of the women did not know it had been made.

P2 · Why do women stop taking it?

Same research question as P1, from the adherence side.

Design: a cohort of high-risk pregnant women with adherence measured objectively — by platelet function analyser and plasma salicylic acid assay, not by asking them — combined with interviews. Findings: 44% of high-risk women were less than 90% adherent. Women with inadequate adherence had higher rates of both early- and late-onset preeclampsia, intrauterine growth restriction and preterm delivery. The reason the authors emphasise is pill burden and unintentional omission — forgetting — and they call for counselling on reminder strategies. Taking the tablet is a separate failure from being prescribed it.

P3 · How is preeclampsia screened for and prevented in Saudi Arabia?

How is preeclampsia risk screening and prevention implemented in routine antenatal care in Saudi Arabia, and what is the local burden of disease?

Design: systematic review and meta-analysis of 21 Saudi studies published between 1999 and 2024, drawn from PubMed, Web of Science and Cochrane; 15 of the 21 reported prevalence. Findings: pooled prevalence 4.80%, rising to 9.52% in cross-sectional studies. Awareness among women 27%. Maternal outcomes included HELLP syndrome and eclamptic seizures; fetal outcomes included stillbirth, prematurity and growth restriction. Worth noticing: the risk factors this review highlights are serum lead, IgA nephropathy and calcium, magnesium and zinc deficiency — not the standard checklist factors. The Saudi literature has studied unusual exposures rather than how the routine checklist performs here, which is itself a gap our interview can fill.

P4 · What do Saudi women know about preeclampsia?

What is the level of awareness and knowledge of preeclampsia among women in Saudi Arabia?

Design: a survey of Saudi women of reproductive age. Findings: limited knowledge of what preeclampsia is, what its warning signs are, and that it can be present while a woman still feels well. Supports the case that a woman cannot report a warning sign she was never told to watch for. Note on the sample: it was recruited through social media, with a minimum sample size of 385, and most respondents were young (mean age 25.9) and single — so it describes women before pregnancy rather than women already in antenatal care.

P5 · When do women first attend antenatal care?

At what gestational age do women attend their first antenatal visit in Saudi Arabia, and how does late booking affect eligibility for first-trimester screening?

Design: cross-sectional study of 277 women registered at antenatal clinics in primary health care centres in Abha, with late booking defined as after 12 weeks. Findings: mean gestational age at booking 12.2 weeks; 36% booked late. Nulliparous women were 2.28 times more likely to book late (95% CI 1.20–4.31, $p = 0.011$), and this was the only significant predictor — age, being a housewife, previous abortion and previous obstetric loss were all non-significant. This is the most useful finding we have: combined screening runs at 11–14 weeks, so more than a third of women arrive with the window already closing, and the group arriving latest is nulliparous, itself a risk factor on the checklist.

P6 · How often is preeclampsia found before symptoms appear?

How often is preeclampsia detected at a routine antenatal visit before symptoms appear, compared with detection after the woman presents with symptoms?

Macedonian Medical Review (2019). Design: retrospective questionnaire study of all women at 37 weeks or more admitted to the Special Hospital for Obstetrics and Gynecology “Mother Teresa” over a two-week period in February 2018. Findings: 22% of women who attended regular antenatal check-ups had no blood pressure measurement taken by their obstetrician at all, and the diagnosis was established early — before symptoms began — in only 16.7% of cases. Conclusion: obstetricians’ awareness of pre-symptomatic detection is low. Caveat: this is one hospital in North Macedonia over two weeks, so it is the weakest of these seven for generalising to Saudi practice. Use it to show the problem exists and is documented, and let the doctor interview supply the local figure.

P7 · Can routine records alone predict preeclampsia?

Which maternal history and routinely collected clinical variables predict preeclampsia in models that do not use biomarkers or Doppler, and how complete are routine antenatal records for this purpose?

Design: a digital phenotyping algorithm curated 108,557 pregnancies from electronic medical records across the Mount Sinai Health System; models were trained on 60,879 pregnancies and validated in two independent cohorts. Findings: areas under the curve of 0.92 at 37 gestational weeks, 0.82 intrapartum and 0.89 postpartum. The models recovered the known risk factors — blood pressure, weight and maternal age — and added complete-blood-count characteristics for the antepartum period. Modelling the trajectory across pregnancy outperformed a single-point assessment, which is the principle BADIRA is built on. Important caveat: the 0.92 figure is at 37 weeks, which is late pregnancy, not the 11–16 week window we are aiming at. This paper proves that routinely collected records carry real signal; it does not prove they carry it early. Do not quote 0.92 as if it supported early prediction.

Part A — Questions for the doctor

About how preeclampsia is prevented and found here. Every answer is a fact about practice.

A1 · The first visit and the record

#	Question	Where it comes from, and why it matters	Answer as
1	Around what gestational week do women usually attend their first antenatal visit here?	P5 found a mean booking of 12.2 weeks in Saudi Arabia, with 36% after 12 weeks. Combined screening runs at 11–14 weeks and aspirin must start before 16 (Rolnik et al., 2017; ACOG & SMFM, 2021). This measures whether the window is open here.	Number — week
2	What information and measurements are routinely collected at the first antenatal visit?	P7 found blood pressure, weight and maternal age among the most influential variables in models built from routine records. We need the list actually collected, not the published one.	Open response → interviewer tick list
3	Where is this information recorded, and can you access the woman's previous history, blood-pressure readings and test results in one record? If she changes clinics or hospitals, is that information still available to you?	P7 shows that routine records can support prediction when previous information is available over time. This checks whether the woman's history, blood-pressure readings and test results are actually accessible together.	Paper / electronic / mixed + description of accessibility

A2 · Finding the women at risk

#	Question	Where it comes from, and why it matters	Answer as
4	How do you identify that a woman is at increased risk of preeclampsia? Do you use a formal checklist, clinical judgement, an electronic tool, or a combination?	P3 recommends early risk identification as part of Saudi antenatal care. This asks whether that is done with a checklist, by judgement, or not at all.	Open response + classification afterward
5	Which risk factors do you routinely check when assessing a woman's risk of preeclampsia?	The published checklist has six high-risk and nine moderate-risk factors (SMFM, 2020; SMFM, 2026). The difference between that list and the one used here is the gap in the record.	Open response → mapped afterward to SMFM list
6	In the first trimester, do you use routine blood pressure only, or a standardized MAP protocol for preeclampsia screening? If MAP is used, how and when is it measured?	Report 1 shows that adding MAP to maternal factors improves prediction without requiring Doppler or PIGF. This checks whether standardized MAP screening is actually used in local practice.	Routine BP / standardized MAP / both + method + gestational week

#	Question	Where it comes from, and why it matters	Answer as
7	Is uterine artery Doppler used here at 11–14 weeks specifically for preeclampsia risk screening? Is it routine for all women, only selected high-risk women, or not routinely used?	Clinical guideline / FMF literature (O’Gorman et al., 2017): uterine artery Doppler is part of combined first-trimester screening. This checks whether it is available and routinely used locally.	Routine / selected cases / unavailable
8	Is PIGF testing available here for preeclampsia risk assessment? When is it used, is it routine or only for selected cases, and approximately how long does the result take?	O’Gorman et al. (2017): PIGF is part of combined screening. This checks local availability and use.	Available yes/no + routine/selected + turnaround time

A3 · Prevention

#	Question	Where it comes from, and why it matters	Answer as
9	When a woman is identified as being at increased risk of preeclampsia, what preventive management do you usually start? If aspirin is prescribed, at what gestational week and dose?	Low-dose aspirin is the only preventive treatment in the guidelines, from 12 weeks and ideally before 16 (ACOG & SMFM, 2021), and it cut preterm preeclampsia by about 62% in the ASPRE trial (Rolnik et al., 2017).	Management + aspirin yes/no + week + dose
10	When aspirin is indicated, is it usually started before 16 weeks in your practice? What commonly causes a delay or prevents it from being started?	P1 found that aspirin is underused even among women who meet the criteria, because risk is not identified and the recommendation is not clearly communicated. This measures that locally.	Open text / reasons
11	Once aspirin has been started, what are the common reasons women miss doses, take it irregularly, or stop it?	P2 found 44% of high-risk women were less than 90% adherent, and documents the reasons. We are asking whether the same reasons appear here.	Open text

A4 · Finding the disease

#	Question	Where it comes from, and why it matters	Answer as
12	How often is blood pressure measured during pregnancy, and at which visits? Is it also monitored at home for some women?	P6 found that 22% of women attending regular check-ups had no blood pressure measured at all. Diagnosis needs two readings at least four hours apart after 20 weeks (ACOG, 2020), and a trajectory needs more than that.	Visits/frequency + clinic/home
13	When you suspect that a woman may have developed preeclampsia, can you walk us through what you do next, from the first abnormal finding until the diagnosis is confirmed or excluded?	Diagnosis requires new hypertension after 20 weeks plus proteinuria or organ dysfunction (ACOG, 2020). NICE and ISSHP differ in detail (NICE, 2019; Magee et al., 2022), so the local sequence must be recorded.	Clinical pathway + tick list afterward
14	Is preeclampsia more often first identified during routine monitoring before the woman reports symptoms, or after she presents with symptoms? If your clinic has actual figures for this, approximately what proportion fall into each group? What usually triggers the first suspicion?	P6 reports that only 16.7% of cases were diagnosed before symptoms began. This is the premise of the whole project, put directly to a clinician.	Routine monitoring / symptoms / mixed + optional recorded percentage + trigger

#	Question	Where it comes from, and why it matters	Answer as
15	Thinking about real cases you have managed, where do you most often see missed opportunities for earlier recognition of preeclampsia risk? Can you give us an anonymized example from a recent case?	P3 and P6 show gaps in early risk identification and pre-symptomatic detection. This asks where those missed opportunities actually occur in local practice and whether they match the literature.	Open text + anonymized case example

Qualitative or quantitative — for the doctor

Mixed, mainly qualitative with some structured factual answers.

The interview collects factual details such as gestational week, aspirin dose, availability of tests, and type of record system, but it also asks the doctor to explain how screening, prevention, and diagnosis actually happen in practice. Numerical answers should only be recorded when they are known from clinic records or routine practice, rather than estimated from memory.

Part B — Questions for the patient

About what actually happened during her pregnancy. The questions focus on real events, timing, communication, monitoring, and care.

B1 · The start of the pregnancy

#	Question	Where it comes from, and why it matters	Answer as
1	In which week of pregnancy did you go for your first antenatal visit?	P5 found a mean of 12.2 weeks and 36% booking after 12 weeks. Everything preventive must happen before 16 (ACOG & SMFM, 2021). Her week places her in or outside that window.	Number — week
2	What information did they ask you about at your first visit, such as previous pregnancies, medical conditions, family history, or your weight?	These are the checklist items (SMFM, 2020). P7 shows these same variables carry the prediction. If she was never asked, the field is empty in her record.	Open response + tick list

B2 · Risk communication and prevention

#	Question	Where it comes from, and why it matters	Answer as
3	Before or during this pregnancy, had anyone explained preeclampsia to you or its warning signs? Where did that information come from?	P4 found limited awareness of preeclampsia among Saudi women. This checks what information the woman actually received.	Yes/no + source + what she was told

#	Question	Where it comes from, and why it matters	Answer as
4	Before any problem was diagnosed, did anyone tell you that you had a higher risk of preeclampsia or a pregnancy-related blood-pressure problem? If yes, when and why?	P1 found that failure to communicate the risk is one of the two reasons aspirin is underused. This asks whether the communication happened at all.	Yes/no + week + reason given
5	Were you prescribed aspirin or any medicine to reduce the risk of preeclampsia? If yes, when did you start it and what reason did the doctor give you?	P1 found that some eligible women were not advised to take aspirin or were unaware of the recommendation. This checks whether it was prescribed and understood. .	Yes/no + week + medicine + reason
6	If aspirin was prescribed, did you take it regularly? If you missed doses or stopped, what happened?	P2 found 44% of high-risk women were less than 90% adherent. Prescribing and taking are two different events, and only this question separates them.	Regular / irregular / stopped + reason

B3 · Monitoring and detection

#	Question	Where it comes from, and why it matters	Answer as
7	How often was your blood pressure measured during pregnancy, and was it measured only at the clinic or also at home?	P6 found 22% of women at regular check-ups had no blood pressure taken. P7 shows the trajectory of blood pressure across pregnancy predicts better than a single reading — so the count of readings matters.	Frequency + clinic/home
8	When did you first notice that something might be wrong, and what did you notice?	Preeclampsia develops after 20 weeks (ACOG, 2020). Her week, set against her booking week from Q2, shows how much of the preventive window passed before anything was noticed.	Week + open description
9	When you reported those symptoms or concerns, what were you told, and what happened next? Were any tests done? Were you sent home, followed up, or admitted?	P6 looks at delayed versus early detection, while clinical guidelines describe the assessment that follows suspected preeclampsia. This checks the real sequence of care.	Open timeline + tests/actions

B4 · Information continuity

#	Question	Where it comes from, and why it matters	Answer as
12	During your pregnancy, were your previous visits, blood-pressure readings, test results, and medical history available to the doctors, or did you have to repeat or bring this information again? If you changed clinic or hospital, was your previous information still available?	P7 shows that prediction depends on having routine information available over time. This checks whether the woman's records were complete and connected across care.	Available / partly available / not available + example

Qualitative or quantitative — for the patient

Mixed, mainly qualitative with some structured factual answers.

The patient interview is designed to reconstruct the sequence of what actually happened during her pregnancy. Qualitative answers explain how risk was communicated, what symptoms were experienced, what response followed, and whether information was available across visits. Structured answers such as gestational week, aspirin use, and blood-pressure monitoring provide time points that allow the experience to be compared with the literature and clinical guidelines.

The analysis will use a timeline and theme-based comparison to identify where care matched the expected pathway and where gaps occurred, such as delayed booking, missed monitoring, unclear risk communication, delayed prevention, or fragmented records.

Putting the two together

The purpose of using both interviews is to compare published evidence with actual clinical practice and a real patient experience.

What to compare	Literature says	Doctor says	Patient says
When care starts	Early antenatal booking is important for first-trimester screening	Most women book at week X	I booked at week Y
Risk assessment	Maternal history and risk factors should be assessed early	We ask about these risk factors	They asked me about these
Records	Longitudinal information is useful for assessing risk over time	Previous information is / is not accessible	My previous results were / were not available
Prevention	Aspirin should be started early when indicated	We usually prescribe at week X, dose X	I was given / not given it at week Y
Adherence	Some women do not take aspirin consistently	Common reasons are X	I missed/stopped it because X
Monitoring	Repeated BP measurements are important	BP is measured at these visits	It was taken X times, at these visits
Advanced screening	FMF may use MAP, UtA Doppler and PIGF	These are / are not routinely available	These tests were / were not done
Detection	PE may be detected during monitoring or after symptoms	It is usually found at this stage	I noticed it in week X and was told this

What to compare	Literature says	Doctor says	Patient says
Missed opportunities	Literature reports gaps in early identification and prevention	Common gaps are X	My experience included X

Where the literature, doctor, and patient findings agree, the published problem is also seen in local practice. Where they differ, this may reveal a local practice gap or a different Saudi care pathway. Findings that appear only in the interviews can identify new real-world issues not captured by the literature. These comparisons will be used to confirm, correct, or expand Report 1.

References

From the literature search

- P1** Krishnamurti, T., et al. (2021). Use of a smartphone app to explore potential underuse of prophylactic aspirin for preeclampsia. JAMA Network Open, 4(10), e2130804. <https://jamanetwork.com/journals/jamanetworkopen/fullarticle/2785604>
- Free full text of the same paper: <https://pmc.ncbi.nlm.nih.gov/articles/PMC8556626/>
- P2** Shanmugalingam, R., et al. (2020). Factors that influence adherence to aspirin therapy in the prevention of preeclampsia amongst high-risk pregnant women: a mixed-method analysis. PLoS ONE. <https://pmc.ncbi.nlm.nih.gov/articles/PMC7046289/>
- P3** Preeclampsia in Saudi Arabia: a systematic review and meta-analysis of prevalence, risk factors, awareness and outcomes (2025). <https://pmc.ncbi.nlm.nih.gov/articles/PMC12790343/>
- P4** Awareness and Knowledge of Pre-eclampsia Among Saudi Women of Reproductive Age. Cureus (2023), 15(11), e49233. <https://pmc.ncbi.nlm.nih.gov/articles/PMC10739410/>
- P5** Siddiqui, A. F. (2016). Late antenatal booking and its predictors among mothers attending primary health care centers in Abha, Saudi Arabia. Rawal Medical Journal, 41(1), 72–76. <https://www.rmj.org.pk/?mno=182812>
- P6** Hypertensive disorders in pregnancy — physicians' awareness for early detection. Macedonian Medical Review, 2019. <https://repository.ukim.mk/handle/20.500.12188/9112>
- P7** Improving preeclampsia risk prediction by modeling pregnancy trajectories from routinely collected electronic medical record data. npj Digital Medicine (2022). <https://pmc.ncbi.nlm.nih.gov/articles/PMC9170686/>

Clinical guidelines and trials

- American College of Obstetricians and Gynecologists. (2020). Gestational hypertension and preeclampsia. ACOG Practice Bulletin No. 222. Obstetrics & Gynecology, 135(6), e237–e260.
- American College of Obstetricians and Gynecologists & Society for Maternal-Fetal Medicine. (2021). Practice Advisory: Low-dose aspirin use for the prevention of preeclampsia and related morbidity and mortality.

- Magee, L. A., Brown, M. A., Hall, D. R., et al. (2022). The 2021 ISSHP classification, diagnosis and management recommendations for international practice. *Pregnancy Hypertension*, 27, 148–169.
- National Institute for Health and Care Excellence. (2019). Hypertension in pregnancy: diagnosis and management. NICE guideline NG133.
- O’Gorman, N., Wright, D., Poon, L. C., et al. (2017). Accuracy of competing-risks model in screening for pre-eclampsia by maternal factors and biomarkers at 11–13 weeks’ gestation. *Ultrasound in Obstetrics & Gynecology*, 49(6), 751–755.
- Rolnik, D. L., Wright, D., Poon, L. C., et al. (2017). Aspirin versus placebo in pregnancies at high risk for preterm preeclampsia. *New England Journal of Medicine*, 377(7), 613–622.
- Society for Maternal-Fetal Medicine. (2020, updated 2026). Special Statement: Checklists for preeclampsia risk-factor screening to guide recommendations for prophylactic low-dose aspirin.